1. Evaluate $\lim_{x\to 0} \dfrac{e^{2x} - \cos x - 2\sin x}{x^2}$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{5}{2}$

$$\lim_{x\to 0} \frac{e^{2x} - \cos x - 2\sin x}{x^2} \overset{\text{l'Hôpital's rule}}{=} \lim_{x\to 0} \frac{2e^{2x} + \sin x - 2\cos x}{2x}$$

$$\overset{\text{l'Hôpital's rule}}{=} \lim_{x\to 0} \frac{4e^{2x} + \cos x + 2\sin x}{2} = \frac{5}{2}$$

2. Let C be the curve defined by $x^2 - 7y^2 + y^3 = 6xy$. Find **all** points (x, y) on C where the tangent line is horizontal.

(48, 16)

$$x^2 - 7y^2 + y^3 = 6xy$$

Differentiate implicitly wrt x:

$$2x - 7 \cdot 2y \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}$$

Group terms: $\frac{dy}{dx}(3y^2 - 14y - 6x) = 6y - 2x$

i.e. $\frac{dy}{dx} = \frac{6y - 2x}{3y^2 - 14y - 6x}$

For horizontal tangent, need $\frac{dy}{dx} = 0$

get $6y - 2x = 0$ i.e. $x = 3y$

Subst. $x = 3y$ into orig. eqn.

get $(3y)^2 - 7y^2 + y^3 = 6(3y)y$

i.e. $9y^2 - 7y^2 + y^3 = 18y^2$

i.e. $y^2(y - 16) = 0$

i.e. $y = 0$ or $y = 16$

corresp. $x = 0$ $x = 48$

i.e. $(0, 0)$ or $(48, 16)$

↑ is a singular point of C tangent line is not defined

3. Let $x = x(t)$ and $y = y(t)$ be functions of t, related by the equation

$$2x^3 + 3y^2 + xy = 6.$$

Given that $x(0) = 1$, $y(0) = 1$ and $x'(0) = 4$, find $y'(0)$.

Give the answer as an integer or a fraction in lowest terms.

-4

Start from $2x^3 + 3y^2 + xy = 6$

Diff. wrt t: $6x^2x' + 6yy' + \underbrace{x'y + xy'}_{\text{pdt rule}} = 0$

Evaluate x, y, x', y' at $t = 0$:

$$6\underbrace{x(0)}_{1}^2\underbrace{x'(0)}_{4} + 6\underbrace{y(0)}_{1}y'(0) + \underbrace{x'(0)}_{4}\underbrace{y(0)}_{1} + \underbrace{x(0)}_{1}y'(0) = 0$$

$$24 + 6y'(0) + 4 + y'(0) = 0 \quad \Rightarrow \quad y'(0) = \frac{-28}{7} = -4$$

4. Evaluate $\displaystyle\int_0^\infty x\,e^{-x/2}\,dx.$

Give the answer as an integer or a fraction in lowest terms.

4

Int. by parts : $\displaystyle\int_0^\infty \underbrace{x}_{u}\,\underbrace{e^{-x/2}}_{v}\,dx = \Big[-2xe^{-x/2}\Big]_{x=0}^{x=\infty} + 2\int_0^\infty e^{-x/2}\,dx$

$= 0 + 2\Big[-2e^{-x/2}\Big]_{x=0}^{x=\infty}$

$= 2\times(+2) = 4$

5. Consider the function f given by

$$f(x) = \int_{x^2}^{3x} \left(\frac{t}{3} + 1\right) dt.$$

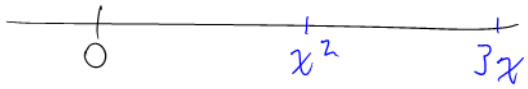
Find $f'(1)$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{10}{3}$

Define function $F(x) := \int_0^x \left(\frac{t}{3} + 1\right) dt$. By FTC, know $F'(x) = \frac{x}{3} + 1$

Our function $f(x)$ is $F(3x) - F(x^2)$



so $f'(x) = F'(3x) \cdot 3 - F'(x^2) \cdot 2x$

$= (x+1) \cdot 3 - \left(\frac{x^2}{3} + 1\right) \cdot 2x$

If $x = 1$, get $f'(1) = (1+1) \cdot 3 - \left(\frac{1^2}{3} + 1\right) \cdot 2 \cdot 1 = 6 - \frac{8}{3} = \frac{10}{3}$

typo !

6. Evaluate $\int_0^{\ln 3} e^x (1+e^x)^2 \, dx$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{56}{3}$

Let $u = 1 + e^x$ Then $du = e^x \, dx$,

$x = 0 \longleftrightarrow u = 2$

$x = \ln 3 \longleftrightarrow u = 4$

So $\int_{x=0}^{x=\ln 3} \underbrace{e^x (1+e^x)^2}_{u^2} dx = \int_{u=2}^{u=4} u^2 \, du = \left[\frac{u^3}{3}\right]_2^4 = \frac{1}{3}(64-8)$

$= \frac{56}{3}$

7. Find the value of the constant c such that the two vectors $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ and $c\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ are orthogonal.

Give the answer as an integer or a fraction in lowest terms.

$\frac{9}{2}$

$2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ and $c\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ are orthogonal

iff $(2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}) \cdot (c\mathbf{i} + 5\mathbf{j} + \mathbf{k}) = 0$

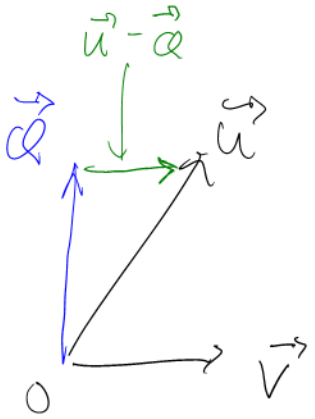
iff $2c - 15 + 6 = 0$

iff $c = \frac{9}{2}$

8. Let $\overrightarrow{U} = 4\mathbf{i}+3\mathbf{j}+4\mathbf{k}$ and $\overrightarrow{V} = \mathbf{i}+2\mathbf{j}+2\mathbf{k}$, and suppose $\overrightarrow{Q}$ is a vector perpendicular to $\overrightarrow{V}$ such that $\overrightarrow{U}-\overrightarrow{Q}$ is parallel to $\overrightarrow{V}$. Find the vector $\overrightarrow{Q}$.

Give the answer in the form $a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$, where a, b, c are real numbers.

$2i - j + 0k$



Since $\vec{u}-\vec{q}$ is // to $\vec{v}$, can write $\vec{u}-\vec{q} = \lambda\cdot\vec{v}$ for some $\lambda \in \mathbb{R}$

i.e. $\vec{q} = \vec{u} - \lambda\vec{v}$

Since $\vec{q}$ is $\perp$ to $\vec{v}$, must have $\vec{q}\cdot\vec{v} = 0$

so $\vec{u}\cdot\vec{v} - \lambda\vec{v}\cdot\vec{v} = 0$

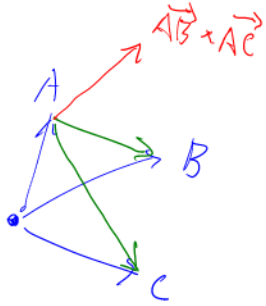
i.e. $\lambda = \dfrac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}} = \dfrac{(4\mathbf{i}+3\mathbf{j}+4\mathbf{k})\cdot(\mathbf{i}+2\mathbf{j}+2\mathbf{k})}{(\mathbf{i}+2\mathbf{j}+2\mathbf{k})\cdot(\mathbf{i}+2\mathbf{j}+2\mathbf{k})} = \dfrac{18}{9} = 2$

so $\vec{q} = \vec{u} - 2\vec{v} = (4\mathbf{i}+3\mathbf{j}+4\mathbf{k}) - 2(\mathbf{i}+2\mathbf{j}+2\mathbf{k}) = 2i - j$

9. Determine the equation of the plane containing the three points $(1,2,-1)$, $(3,1,0)$ and $(-1,0,2)$.

Give the equation in the form $ax+by+cz=1$ where each of a,b,c is an integer or a fraction in lowest terms.

$$\frac{x}{11}+\frac{8y}{11}+\frac{6z}{11}=1$$



Let $A=(1,2,-1)$ Then $\vec{AB}=(2,-1,1)$ are // to plane thru ABC

$B=(3,1,0)$ $\vec{AC}=(-2,-2,3)$

$C=(-1,0,2)$

so $\vec{AB}\times\vec{AC}$ is $\perp$ to plane thru ABC

$$=\det\begin{vmatrix} i & j & k \\ 2 & -1 & 1 \\ -2 & -2 & 3 \end{vmatrix} = -i-8j-6k$$

Plane is $\perp$ to $-i-8j-6k$ and goes thru $A=(1,2,-1)$

so eqn of plane is $\big((x,y,z)-(1,2,-1)\big)\cdot(-i-8j-6k)=0$

i.e. $-x-8y-6z=-1-16+6$

i.e. $\frac{x}{11}+\frac{8y}{11}+\frac{6z}{11}=1$

Alt. Mthd:

Plug into $ax+by+cz=1$

$(x,y,z)=\begin{cases}(1,2,-1)\\(3,1,0)\\(-1,0,2)\end{cases}$

get system of 3 linear eqns in unknowns a,b,c

solve to get answer

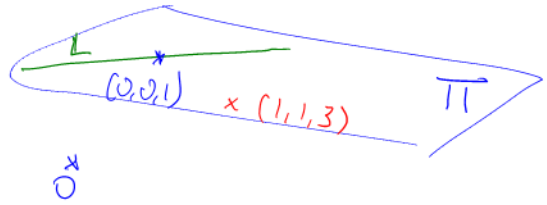
10. Let L be the line with parametric equations

$$x = t, \quad y = 2t, \quad z = 1.$$

Let Π be the plane that contains L and passes through the point $(1, 1, 3)$. Find an equation of the plane Π.

Give the answer in the form $z = ax + by + c$, where a, b, c are real constants.

$z = 4x - 2y + 1$



Line L is given by $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ is // to plane Π

Another vector // to plane Π is given by $\begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

so a normal vector to plane Π is $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{vmatrix} i & j & k \\ 1 & 2 & 0 \\ 1 & 1 & 2 \end{vmatrix} = 4i - 2j - k$

so eqn of plane is $\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \right) \cdot \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix} = 0$

ie. $4x - 2y - z = 4 - 2 - 3 = -1$ ie. $z = 4x - 2y + 1$

11. The plane defined by the equation $2x - y + 3z = 10$ intersects the line parallel to the vector $\langle 1, 2, 1 \rangle$ and passing through the point $(1, 1, 1)$. Determine the point of the intersection.

$$\begin{pmatrix} 3 \\ 5 \\ 3 \end{pmatrix}$$

Plane Π, eqn. $2x - y + 3z = 10$

Line L, eqn $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ where $t \in \mathbb{R}$ is a parameter

At pt of intersection, get $2(\underbrace{1+t}_{x}) - (\underbrace{1+2t}_{y}) + 3(\underbrace{1+t}_{z}) = 10$

ie. $3t = 6$ ie. $t = 2$

so pt of intersection has coords $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 3 \end{pmatrix}$

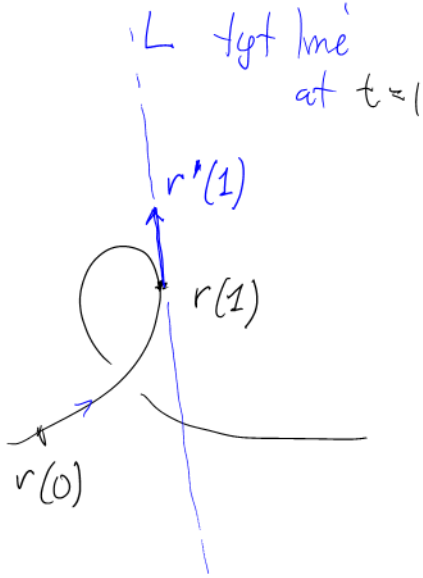
12. Let C be the curve in the xyz-space with parametrization given by

$$\mathbf{r}(t) = (3 - t^2)\mathbf{i} + t^3\mathbf{j} + (4t - 1)\mathbf{k}.$$

Let L be the tangent line to C at the point where $t = 1$. Find the point at which L intersects the yz-plane.

Give the answer in the form $(0, b, c)$, where b, c are real numbers.

$$\begin{pmatrix} 0 \\ 4 \\ 7 \end{pmatrix}$$



Diff. $\mathbf{r}(t) = \begin{pmatrix} 3-t^2 \\ t^3 \\ 4t-1 \end{pmatrix}$ wrt t, get $\mathbf{r}'(t) = \begin{pmatrix} -2t \\ 3t^2 \\ 4 \end{pmatrix}$

At $t=1$, get $\mathbf{r}(1) = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\mathbf{r}'(1) = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$

so L passes thru $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and // to $\begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$

so eqn of L is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$

At intersctn of L with yz-plane, get $x = 0$, ie. $2 - 2\lambda = 0$ ie. $\lambda = 1$

so coords of intersctn of L with yz-plane is $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + 1 \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 7 \end{pmatrix}$